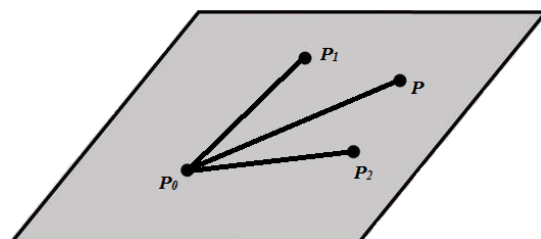


MTH 202 Chapter 1: Three-Dimensional Space & Vectors

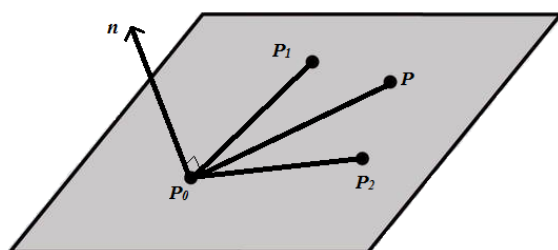
1.6 Planes in 3-Space

We now move to thinking about planes, and we ask ourselves what information do we need in order to determine an equation of a plane? Note that two points determine a line and three non-collinear points in $\mathbb{R}^3$ determine a plane.

Suppose we have a plane being determined by points P_0 , P_1 , & P_2 . Suppose there is another point P with coordinates (x, y, z) . We would like to establish a condition telling us whether P is in the plane determined by P_0 , P_1 , & P_2 .



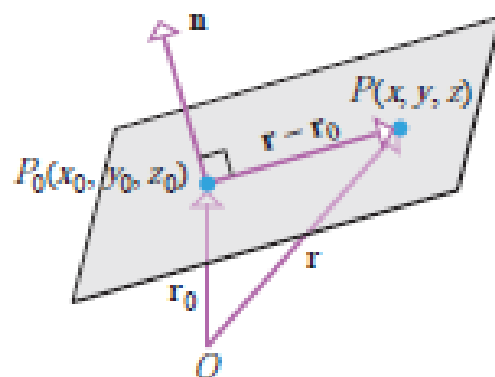
It turns out that one of the simplest ways to establish this condition is to note that the point P with coordinates (x, y, z) will be in the plane only if the vector $\overrightarrow{P_0P}$ is perpendicular to a vector that is perpendicular to the plane.



Definition: We say that a vector is normal to a plane if it is perpendicular to the plane.

Therefore, we establish that P is in the plane if it is the case that $\overrightarrow{P_0P} \cdot \mathbf{n} = 0$

Another way of looking at this would be to think about the vectors formed by the origin and the points P_0 & P . In the picture below we let O represent the origin, $\overrightarrow{OA} = \mathbf{r}_0$, $\overrightarrow{OP} = \mathbf{r}$, and $\mathbf{n}$ is a nonzero vector normal to the plane. We then notice that $\overrightarrow{P_0P} = \mathbf{r} - \mathbf{r}_0$. As stated previously we see that $\mathbf{n}$ and $\overrightarrow{P_0P} = \mathbf{r} - \mathbf{r}_0$ are orthogonal. Thus we have that $\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_0) = 0$.



Terminology: If $\mathbf{r}_0$ is the position vector of a point on a plane and $\mathbf{n}$ is a vector normal to the plane, then $\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_0) = 0$ is called the vector equation of the plane.

Note: The vector equation of a plane can be used beyond three dimensions, and it can be used in two dimensions to yield the equation of a line! In this class, planes in four or more dimensions will be known as hyperplanes.

If we return our focus to 3-space and suppose that $\mathbf{n} = \langle a, b, c \rangle$, $\mathbf{r}_0 = \langle x_0, y_0, z_0 \rangle$, and $\mathbf{r} = \langle x, y, z \rangle$, then clearly we obtain that:

$$\begin{aligned}\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_0) &= 0 \\ \Rightarrow \langle a, b, c \rangle \cdot \langle x - x_0, y - y_0, z - z_0 \rangle &= 0 \\ \Rightarrow a(x - x_0) + b(y - y_0) + c(z - z_0) &= 0.\end{aligned}$$

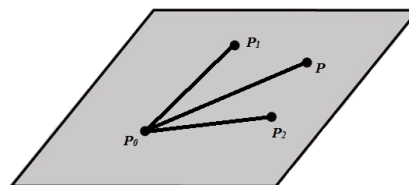
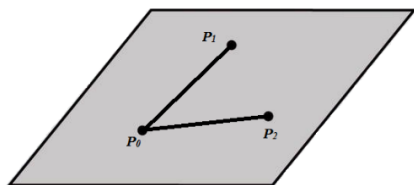
Terminology: The equation $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$ is known as the point-normal form of the equation of a plane through (x_0, y_0, z_0) with normal vector $\mathbf{n} = \langle a, b, c \rangle$.

If we let $d = ax_0 + by_0 + cz_0$ we can rewrite the scalar equation of the plane as $ax + by + cz = d$.

Theorem: If a, b, c , and d are constants (with a, b , and c not *all* zero), then the graph of the equation $ax + by + cz = d$ is a plane that has the vector $\mathbf{n} = \langle a, b, c \rangle$ as a normal vector.

Alternative (yet equivalent) Approaches to Determining the Equation of a Plane (Not Done in Class):

Another possible way of establish a condition telling us whether P is in the plane determined by P_0, P_1 , & P_2 would be to notice that if we create the vectors $\overrightarrow{P_0P_2}$, $\overrightarrow{P_0P_1}$, & $\overrightarrow{P_0P}$ we will determine a parallelepiped which must have some volume. As stated in section 1.4, we will also notice that this volume will be zero precisely when all three vectors are coplanar which means that all four points (P_0, P_1, P_2 , & P) are coplanar. So we could set up and simplify the equation $\overrightarrow{P_0P} \cdot (\overrightarrow{P_0P_2} \times \overrightarrow{P_0P_1}) = 0$



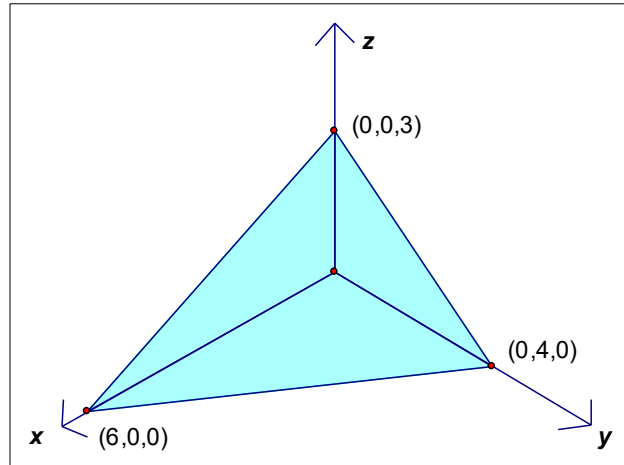
We will notice however that $\overrightarrow{P_0P_2} \times \overrightarrow{P_0P_1}$ simply creates a normal vector to the plane (suppose we call this vector $\mathbf{n}$). Therefore, we can also simplify the plane equation formula to read $\overrightarrow{P_0P} \cdot (\overrightarrow{P_0P_2} \times \overrightarrow{P_0P_1}) = 0 \Rightarrow \overrightarrow{P_0P} \cdot \mathbf{n} = 0$. We have arrived exactly back at where our first approach led us.

Example 1: Find an equation (in the form $ax+by+cz=d$) of the plane through the point $(2,4,-1)$ with normal vector $\langle 2,3,4 \rangle$. Also sketch the portion of this plane which is in the first octant.

Based on what we discussed above we know that an equation of the plane is: $2(x-2)+3(y-4)+4(z+1)=0$.
If we manipulate this equation we obtain: $2x+3y+4z=12$

(Note: A slicker way to find the equation of the plane is to start with $2x+3y+4z=?$ and plug in the coordinates of the point that you know must be on the plane).

We note that the x -intercept of the plane is $(6,0,0)$, the y -intercept is $(0,4,0)$, and the z -intercept is $(0,0,3)$. Thus, a sketch of the portion of the plane in the first octant is:



Example 2: Find an equation of the plane that passes through the points $A: (1,3,3)$, $B: (3,-1,5)$, and $C: (5,2,-1)$.

We must find a vector normal to the desired plane. We know that this plane must contain the vectors: $\overrightarrow{AB} = \langle 2, -4, 2 \rangle$ and $\overrightarrow{AC} = \langle 4, -1, -4 \rangle$. Since these two vectors, are not parallel (we know this because they are not scalar multiples of each other) we know that a vector normal to both these vectors will be normal to the plane. We can find a vector normal to both these vectors by taking the cross product:

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -4 & 2 \\ 4 & -1 & -4 \end{vmatrix} = \langle 18, 16, 14 \rangle.$$

Now, we know that $\langle 18, 16, 14 \rangle$ is normal to the plane and $(1,3,3)$ is on the plane.

So, an equation for the plane is $18x+16y+14z=108 \Rightarrow 9x+8y+7z=54$

Definition: Two planes are parallel if their normal vectors are parallel. If two planes are not parallel, then they intersect in a straight line and the angle between the two planes is defined to be the acute or right angle between their normal vectors (one must choose the normal vectors so that their dot product is nonnegative).

Example 3: Consider the planes given by $2x - 2y + z = 4$ and $x + 4y + z = 1$.

a) Determine whether the planes parallel.

We quickly observe that $2x - 2y + z = 4$ has a normal vector of $\mathbf{n}_1 = \langle 2, -2, 1 \rangle$ and

$x + 4y + z = 1$ has a normal vector of $\mathbf{n}_2 = \langle 1, 4, 1 \rangle$

Clearly these vectors are *not* scalar multiples of each other and therefore these vectors are not parallel. Thus, the two planes are *not* parallel.

b) Find the angle (rounded to the nearest hundredth) between the planes.

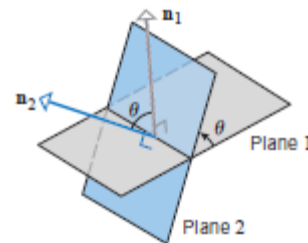
We stated above that $\mathbf{n}_1 = \langle 2, -2, 1 \rangle$ and $\mathbf{n}_2 = \langle 1, 4, 1 \rangle$. If we can find the angle between these normal vectors we can determine the angle between the

intersecting planes. We plan to use the formula $\cos(\theta) = \frac{\mathbf{n}_1 \cdot \mathbf{n}_2}{\|\mathbf{n}_1\| \|\mathbf{n}_2\|}$, however

we note that the dot product of our two vectors is negative which means that solving for θ will produce an obtuse angle. We also prefer to use an acute

angle when describing the angle between two planes. Therefore we plan to write $\cos(\theta) = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{\|\mathbf{n}_1\| \|\mathbf{n}_2\|}$ and we calculate:

$$\cos(\theta) = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{\|\mathbf{n}_1\| \|\mathbf{n}_2\|} = \frac{5}{3\sqrt{18}} \Rightarrow \theta = \cos^{-1}\left(\frac{5}{3\sqrt{18}}\right) \approx 1.17 \approx 66.9^\circ$$



Note: If time permits, the instructor will show a 3 dimensional picture.

c) Write a parametric equation for the line of intersection for the planes.

We need a point and a direction vector $\mathbf{v}$ to establish the equation of this line.

Clearly $\mathbf{v}$ must be perpendicular to $\mathbf{n}_1$ and $\mathbf{n}_2$. So we compute $\mathbf{v} = \mathbf{n}_1 \times \mathbf{n}_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -2 & 1 \\ 1 & 4 & 1 \end{vmatrix} = -6\mathbf{i} - \mathbf{j} + 10\mathbf{k}$

To find a point on the line we first observe that since $\mathbf{v} \cdot \mathbf{k} = 10 \neq 0$ we know that the $\mathbf{v}$ is not perpendicular to $\mathbf{k}$ and thus it is angled and must intersect the xy -plane which is $z = 0$. If we substitute $z = 0$ into our equations we get: $\begin{cases} 2x - 2y = 4 \\ x + 4y = 1 \end{cases} \rightarrow x = \frac{9}{5} \text{ \& } y = -\frac{1}{5}$. Thus, $\left(\frac{9}{5}, -\frac{1}{5}, 0\right)$ is a point on the

line. Therefore the equation of the line can be written as: $\begin{cases} x = -6t + 9/5 \\ y = -t - 1/5 \\ z = 10t \end{cases}$

Example 4: Find the exact distance between the point $A: (-1, 1, 2)$ and the plane given by: $x + 4y + z = 1$.

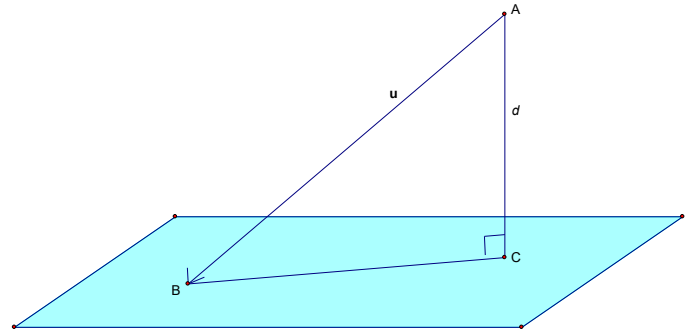
We know that $B: (1, 0, 0)$ is a point on the plane of interest, and $\mathbf{n} = \langle 1, 4, 1 \rangle$ is a vector normal to the plane of interest. A sketch of the situation is:

We wish to find $d = \|\overrightarrow{AC}\|$. We know that $\overrightarrow{AC}$ is normal

to the plane. Thus, if $\mathbf{n}$ is positioned so that its initial point is at A it would run in the exact same or exact opposite direction of $\overrightarrow{AC}$. Also $\mathbf{u} = \overrightarrow{AB} = \langle 2, -1, -2 \rangle$.

We see that the distance is

$$\|\overrightarrow{AC}\| = |\text{comp}_{\mathbf{n}}(\mathbf{u})| = \frac{|\mathbf{u} \cdot \mathbf{n}|}{\|\mathbf{n}\|} = \frac{|-4|}{\sqrt{18}} = \frac{4}{3\sqrt{2}} = \frac{4\sqrt{2}}{6} = \frac{2\sqrt{2}}{3}.$$



Note: It is possible to generalize this process given the plane $ax + by + cz - d = 0$, the point $A: (x_0, y_0, z_0)$, and point on the plane $B: (x_1, y_1, z_1)$. Here we know that $\mathbf{n} = \langle a, b, c \rangle$ and

$$\begin{aligned} d = \|\overrightarrow{AC}\| &= |\text{comp}_{\mathbf{n}}(\mathbf{u})| = \frac{|\mathbf{u} \cdot \mathbf{n}|}{\|\mathbf{n}\|} = \frac{|a(x_0 - x_1) + b(y_0 - y_1) + c(z_0 - z_1)|}{\sqrt{a^2 + b^2 + c^2}} \\ &= \frac{|ax_0 + by_0 + cz_0 - (ax_1 + by_1 + cz_1)|}{\sqrt{a^2 + b^2 + c^2}} = \frac{|ax_0 + by_0 + cz_0 - (-d)|}{\sqrt{a^2 + b^2 + c^2}} \\ &= \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}} \end{aligned}$$